Student dropout warnings

English presentation script

Main talk: slides 1-8 | 599 words in the spoken blocks | 4 suggested speakers

The spoken blocks take about 6.0 minutes at 100 words per minute, before pauses. Their main transitions are included. Rehearse aloud; the appendices are preparation material, not extra speech.

Speaker 1 | Slides 1-2

Slide 1: the question

Our question is whether information known at enrolment can identify later recorded dropout, and who the same warning rule misses. We are interested in student support. We first define the outcome, then evaluate overall prediction, and finally check how errors differ between groups. We start with the label, because it determines what a warning can mean.

Slide 2: the outcome definition

We used 4,424 records from one Portuguese institution. We checked the original paper because dropout can mean different things. Here it includes changing course or institution, so it does not always mean leaving higher education. The outcomes were recorded at the normal end of the course, usually three years. We retained students still enrolled. In our binary analysis they have no recorded dropout at that point; they are not assumed to have graduated. These choices define what our predictions can mean.

Speaker 2 | Slides 3-4

Slide 3: what the entry profiles show

These plots use training records only. The dropout group had a higher median age, but ages and admission grades still overlapped. Quiz 7 pointed us to Data to Viz. Its boxplot advice reminded us to show group sizes, so we added the counts below each outcome. We kept the boxes to compare typical values and spread. These observed patterns are associations, not causes.

Slide 4: inputs and evaluation design

We use age, admission grade and course together, treating course as a category. Our model assumes additive effects on the log-odds scale, without interactions. The regression lectures help us review these assumptions; logistic regression needed further learning. We exclude later grades and financial fields with uncertain timing, and reserve recorded gender for the audit. Training records fit the model, validation records chose one threshold, and test records evaluated the frozen rule. This keeps a small model and separates its decisions. Next, we explain how probabilities become warnings.

Speaker 3 | Slides 5-6

Slide 5: the warning threshold

We wanted to find at least seventy per cent of dropout records in validation data while limiting false warnings. This illustrative target selected a common threshold of 0.31. It found 200 of 284 validation dropouts, about seventy per cent. We then froze the rule. The next slide checks test sensitivity and precision, and the cost in false warnings.

Slide 6: the actual test results

Sensitivity tells us how many dropout records we find; precision tells us how many warnings match a dropout. Lowering the threshold found 107 more dropout records, but produced 177 more false warnings. Our chosen rule found 187 of 285 dropouts: about sixty-six per cent. Only 187 of its 409 warnings matched dropout: about forty-six per cent. Always giving no warning scores higher on accuracy, but finds no dropouts. That is why accuracy alone does not answer our question. We next ask who our rule misses.

Speaker 4 | Slides 7-8

Slide 7: whom the rule misses

We focus first on missed dropouts, because a missed warning could mean a missed offer of support. The rule missed 64 of 144 female dropout records and 34 of 141 male records: forty-four versus twenty-four per cent. The gap was about twenty percentage points, with a conditional interval of roughly ten to thirty-one points. Male students had a higher false-warning rate. The other rows check who receives warnings and recorded outcomes within both prediction groups. These are independence, separation and sufficiency. Different recorded dropout rates complicate those comparisons; the gaps alone do not establish discrimination.

Slide 8: the conclusion and next step

Our enrolment model found about two-thirds of recorded dropouts, with many false warnings and different error rates across groups. Before real use, we would check departure reasons, support capacity and performance on later cohorts at the intended institution. Any warning should lead to human review, with a way to correct records. We have not tested whether acting on warnings prevents dropout. That needs a separate evaluation of the support offered. Thank you.

Preparation notes

Use these explanations selectively when discussing the figures or answering questions. They are not additions to the timed talk.

Slide 2: why we presented the figure this way

We wanted the audience to see all three recorded outcomes, including students still enrolled. After checking the axis advice, we kept the existing bars: they already started at zero and gave both counts and percentages. We did not change a chart that already supported this comparison. Its colours distinguish categories; their order does not rank students.

Slide 3: why we presented the figure this way

We wanted a quick comparison of typical values and spread. Quiz 7 recommended Data to Viz, which led us to review these figures. The boxplot page pointed out that boxes can hide group sizes and distribution shapes. We kept boxplots for this limited comparison and added the training counts: Graduate 1,324, Enrolled 476 and Dropout 852. Equal-width boxes should not suggest equal numbers of students. We did not add thousands of raw dots to the main slide, and we do not claim the boxes reveal every distribution feature. The two panels use different, explicitly labelled units; we compare outcomes within each panel.

Slide 5: why we presented the figure this way

We wanted to show how three rates change as the numerical threshold moves. The axis discussion helped us distinguish this purpose from comparing bar lengths. We kept the lines and the full 0-100% scale. The colour caveats also reminded us to consider readers who cannot distinguish our colours. We gave the three curves different line patterns, so colour is no longer the only way to follow them. These are observed validation rates, not confidence bands; the selected threshold is still 0.31.

Slide 7: why we presented the figure this way

We wanted to compare error rates, but a group's total is not the denominator of every rate. The sample-size caveat prompted us to put the relevant n beside every point. For missed dropouts, those denominators are 144 female and 141 male dropout records, rather than all 575 and 314 students. We kept points with intervals because we are displaying binary rates and sampling uncertainty. The error-bar page stresses that bars can mean different things, so we explicitly identify ours as pointwise 95% percentile bootstrap intervals, conditional on the fixed model and threshold. They are not standard deviations or a guarantee about a future cohort. We also added circle and triangle markers. To compare the groups directly, we use the gap intervals in backup slide 12 rather than treating overlap of two separate intervals as a formal test.

Our reasons for key choices

Preparation material: use the reasoning relevant to a question; do not add every paragraph to the timed talk.

Slide 2: our reasoning

We kept Enrolled because our question concerns recorded dropout at the stated endpoint. Removing those students would change the population. We therefore call them no recorded dropout at that point, without claiming they graduated.

Slide 4: our reasoning

We want an enrolment warning, so a useful predictor must be known at enrolment. Later grades may predict well but answer a different question. We therefore keep a small set of defensible inputs and leave timing-uncertain fields out.

Slide 5: our reasoning

We use the seventy per cent validation target to make our trade-off explicit. It is our illustrative choice, not university policy. Once it selects a threshold, we keep that rule fixed so the test set can still provide a separate evaluation.

Slide 6: our reasoning

We cannot judge this warning by accuracy alone. Always predicting no dropout already gives 67.9% accuracy, while finding none of the recorded dropouts. We therefore report who is found and how many warnings are false, with separate denominators.

Slide 7: our reasoning

We compare missed-dropout rates within actual dropouts, because the groups have different sizes. The relevant comparison is 64/144 against 34/141. We also keep false warnings and both prediction strata, so our fairness discussion covers more than one favourable statistic.

Questions to practise

How do the newer regression lectures connect to the project?

Introducing linear models, pages 48-54, explains reference groups and dummy variables. This helps us explain why course is categorical. Model checking and multiple linear regression, pages 24-41, explains multiple predictors, conditional interpretations and interactions; pages 57-63 motivates parsimony. We reviewed these connections on 11 November after our original evaluation. Logistic regression, classification scores and the validation rule still require supplementary learning.

What does no interaction mean in our model?

We use dropout ~ age10 + admission10 + course. The age coefficient is common across courses on the log-odds scale, as is the admission-grade coefficient. A term such as age10:course would allow the age association to differ by course. The corresponding probability curves need not be parallel even without that interaction. We have not fitted or demonstrated an improvement from interaction terms.

What checks have we actually performed, and what remains assumed?

We checked factor coding and timing, compared against a baseline, kept evaluation partitions separate and examined held-out scores, error counts, group rates and the descriptive calibration figure. Calibration compares average predicted risk with observed dropout proportions in common bands. Coarse bins, small samples and the lack of intervals limit that display. These checks do not establish the correctness of linear age/grade terms or rule out interactions. Those remain explicit assumptions.

Why not use the lecture's adjusted R-squared or ordinary residual rule?

The lecture's ordinary and adjusted R-squared compare OLS models for continuous responses; neither is our classification accuracy. For the binary dropout endpoint we report AUC for ranking, Brier score and calibration for probabilities, and sensitivity, precision and error rates for the warning rule. Binary raw residuals have a structured form, so we do not require the OLS residual cloud shown in the lecture. No single one of our scores establishes suitability or fairness.

Is log-transforming price the same as logistic regression?

No. The lecture models log(price), a transformed continuous response. Logistic regression instead models log(p/(1-p)), the log-odds of a binary outcome, then converts that score to a probability. A logistic coefficient is not a percentage-point change in probability. Its exponential is an odds ratio for the stated predictor change, holding the other included predictors fixed.

Why keep a small model, and how could interactions be explored later?

We chose a small set of defensible enrolment inputs so we can explain the timing, coding and assumptions. The lecture's parsimony principle supports requiring evidence before adding complexity; it does not show that our model is optimal. Any later interaction analysis should be specified and evaluated within development data, with small course groups and overfitting considered. We would not repeatedly select models or thresholds using already-viewed test results and present them as a fresh independent evaluation.

Does higher accuracy mean a better warning rule?

Not for this purpose on its own. Always giving no warning is correct for 604 of 889 records (67.9%), but finds none of the 285 recorded dropouts. The chosen rule is correct for 569 records (64.0%), finds 187 dropouts and produces 222 false warnings. Whether that trade-off is acceptable depends on support capacity and the consequences of each error; these data do not establish an optimal policy.

What do the three fairness criteria ask here?

Independence asks whether students in each recorded-gender group receive warnings at the same rate: 218/575 female records (37.9%) versus 191/314 male records (60.8%). Separation checks errors conditional on the actual label: false-warning rates are 138/431 (32.0%) and 84/173 (48.6%); missed-dropout rates are 64/144 (44.4%) and 34/141 (24.1%). Binary Sufficiency checks outcomes within both prediction groups: dropout among warned records is 80/218 (36.7%) versus 107/191 (56.0%); dropout among unwarned records is 64/357 (17.9%) versus 34/123 (27.6%). These are conditional comparisons, not interchangeable measures or a causal verdict.

Does a 20-point gap prove discrimination?

No. The female-minus-male missed-dropout gap is about 20.3 percentage points in this held-out sample. Its interval conditions on the fitted model and threshold. It does not include all model-development uncertainty, account for unavailable cohort structure, or explain why the groups differ. The pattern warrants investigation before any use; it does not establish a causal explanation or identify who needs support.

Which test counts support sensitivity and precision?

We found 187 of the 285 recorded dropouts, giving sensitivity of 65.6%. We issued 409 warnings; 187 matched a dropout record, giving precision of 45.7%. The other 222 warnings were false positives relative to the recorded endpoint. We missed 98 dropout records and correctly left 382 other records unflagged. We use these counts to explain the denominators rather than treating every percentage as accuracy.

Why use this dataset and this question?

The brief lists UCI as a source for alternative datasets. This release has an accessible source, numerical and categorical variables, and enough student records for a manageable project. Our question connects prediction to a concrete support concern. Dataset feasibility still needs the lecturer/course organiser's confirmation.

What exactly does a positive outcome mean?

Recorded Dropout at the source's normal course endpoint. The paper includes course and institution changes. It does not necessarily mean abandoning all higher education, and it is not a direct observation of who needs support.

Why retain students who are still enrolled?

The primary endpoint is recorded dropout by that point. Enrolled students have not been recorded as Dropout then, though they may leave later. The separate resolved-outcome check makes the consequences of excluding them visible; it cannot silently replace the primary population.

Why not use the very predictive second-semester grades?

They were not available at the chosen prediction time. They could support a different, later-stage question, but would not test an enrolment warning.

Why logistic regression, and is it in the supplied slides?

The two newly supplied lectures teach linear regression, including multiple predictors and categorical effects. They do not directly teach binomial logistic regression. We used AI explanations and official R documentation for that extension, then reimplemented and checked the main analysis. We use logistic regression because our response is binary and we need estimated probabilities. Our age and grade terms are additive and linear on the log-odds scale, without interactions.

Why only three predictors?

They have a defensible enrolment interpretation and keep the model explainable. We chose scope over a broad feature search. This does not prove other entry variables are irrelevant or that this model is optimal.

Why three partitions?

We needed extra explanation of held-out evaluation. We learned to give each partition a different job: training fits coefficients, validation chooses the rule, and test estimates performance after both are fixed. We implemented this split in R and kept validation records out of the final fit. We did not use test results to choose the threshold.

Why seventy per cent?

We chose seventy per cent as an illustrative validation sensitivity goal. We learned how a threshold trades missed records against false warnings, then implemented a rule that maximised specificity among the eligible validation thresholds. We do not present our target as university policy or an objectively optimal choice. Real use would require consultation about benefits, costs and capacity.

What does AUC tell us?

We learned that AUC measures ranking rather than threshold-based accuracy. Our primary test AUC is 0.718, with a conditional bootstrap interval of 0.682 to 0.754. We interpret it as how often a randomly selected dropout record receives a higher score than a non-dropout record, counting ties halfway. We do not treat this as proof of calibration or usefulness at a particular threshold.

What does Brier score tell us?

We learned to evaluate the probabilities as well as the binary warnings. We used Brier score, the mean squared difference between predicted probability and the zero/one outcome; lower is better on comparable records. Our primary test value is 0.188, versus 0.218 for the training-prevalence baseline.

Why is female FNR higher even though gender was not an input?

Omitting a recorded attribute does not remove differences carried by other inputs, course composition or underlying outcome patterns. This descriptive audit identifies a difference. It does not isolate its cause or prove intentional discrimination.

Did adding gender solve the difference?

No. In this pre-specified comparison at the same threshold, AUC changed only slightly and the missed-dropout pattern became more unequal. That is why a small aggregate improvement cannot by itself determine a fair support rule. Both sets of results remain visible in the backup.

Can all three fairness criteria hold?

They condition on different denominators and may conflict when groups have different outcome rates and predictions are imperfect. We do not require all three to hold or make an unconditional impossibility claim. Which error matters depends on the use and the people affected.

What uncertainty does the bootstrap cover?

It resamples held-out student records within each recorded-gender group, with the fitted model and threshold held fixed. It describes variation in this conditional evaluation. It does not account for refitting the model, cohort clustering, transport to another institution or the effect of offering support. The intervals are pointwise, not simultaneous fairness certification.

Is this a future-cohort test?

No. The release has no cohort/year field, so our random hold-out mixes historical cohorts. Prospective or chronological validation is an important missing step.

What is still needed from the group?

Review the data definitions and code, obtain the required topic-feasibility confirmation, agree the speakers, practise aloud, attend check-ins, set up actual repository collaboration and complete the separately required report, peer evaluation and submissions. None is claimed completed by the slide files.

What assistance was used?

We asked ChatGPT to explain unfamiliar methods and used its help with initial code and text drafts, source checks and slide formatting. We then rewrote and ran the main R analysis and checked the outputs. R generated the statistical results; we are responsible for our final interpretation.

Figure-design references

Data to Viz: The Boxplot and its pitfalls - https://www.data-to-viz.com/caveat/boxplot.html

Data to Viz: The issue with error bars - https://www.data-to-viz.com/caveat/error_bar.html

Review workflow: Kassis et al. (2026), Scientific Agent Skills. https://doi.org/10.48550/arXiv.2609.00065

Writing review: Sciwrite (Lorena A. Barba), https://github.com/labarba/sciwrite; Anti-defensive Writing, https://github.com/Kiterlin/anti-defensive-writing